

BPHO ROUND 0 · QUESTION PAPER

Drill bank section 2

25 questions · 60 minutes · single answer, 5 options · **no calculator** · no negative marking*Original questions written for this course, in the Round 0 format. Not BPhO material.***Before you start**

- Answer every question. There is no negative marking, so a blank is a wasted mark.
- No calculator. Every answer here is reachable by hand, and the numbers are chosen to divide cleanly.
- Work on paper, then mark yourself against the markscheme PDF.
- The 2025 UK qualifying line was 11 out of 25, so about 44% is the target to beat.

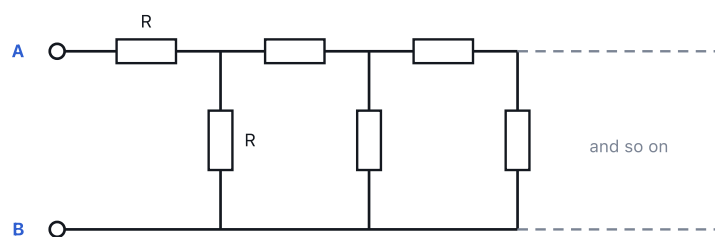
Q1 · Dimensional analysis: the speed of a capillary ripple **A**

Very short ripples on deep water are held in shape by surface tension rather than by gravity. The speed v of such a ripple is found to depend only on the surface tension γ of the water, on its density ρ , and on the wavelength λ . Which expression is dimensionally a speed?

- A** $\sqrt{(\gamma\rho/\lambda)}$ **B** $\sqrt{(\gamma\lambda/\rho)}$
C $\sqrt{(\gamma/\rho\lambda)}$ **D** $\gamma/(\rho\lambda)$
E $\sqrt{(\gamma\rho)}/\lambda$

Q2 · An endless resistor ladder: resistance between the input terminals **H**

The network below repeats the same two-resistor section without end: a resistor R in series, then a resistor R in parallel with whatever follows.

Every resistor in the network has the same resistance R .

What is the resistance measured between the two terminals on the left?

- A** $R(1 + \sqrt{5})/2$ **B** $R(1 + \sqrt{3})$
C $2R$ **D** $R\sqrt{2}$
E $R(3 + \sqrt{5})/2$

Q6 · A water jet striking a wall: the force from a rate of momentum loss **C**

Water of density ρ flows steadily along a pipe of internal cross-sectional area A with speed v . It leaves the pipe and strikes a flat wall head-on, coming to rest against it. What is the average force the water exerts on the wall?

A ρAv

B $2\rho Av^2$

C $\rho Av^2/2$

D $\rho A^2 v$

E ρAv^2

Q7 · Extension of a wire when both the length and the diameter are changed **E**

A wire of length L and diameter d extends by e when a load F is hung on it. A second wire of the same material has length $2L$ and diameter $d/2$. What load is needed to make it extend by the same amount e ?

A $F/8$

B $F/4$

C $F/2$

D $2F$

E F

Q8 · Doppler shift: the frequency heard from a source closing at $v/5$ **F**

A car sounding its horn at a steady frequency travels in a straight line towards a stationary observer at a steady speed of $v/5$, where v is the speed of sound in air. What is the ratio of the frequency the observer hears to the frequency the horn emits?

A $5/4$

B $4/5$

C $6/5$

D $25/16$

E $5/6$

Q9 · A prism at minimum deviation: finding the deviation from the refractive index **G**

A ray of monochromatic light passes through a glass prism of refracting angle 60° and refractive index $\sqrt{2}$. The prism is turned until the deviation is as small as it can be. What is the angle of minimum deviation?

A 45°

B 60°

C 30°

D 15°

E 22.5°

Q10 · Parallel resistors: which one takes the larger share of the power H

A $6\ \Omega$ resistor and a $3\ \Omega$ resistor are connected in parallel across a battery of negligible internal resistance. What fraction of the total power supplied by the battery is dissipated in the $3\ \Omega$ resistor?

A $1/3$

B $2/3$

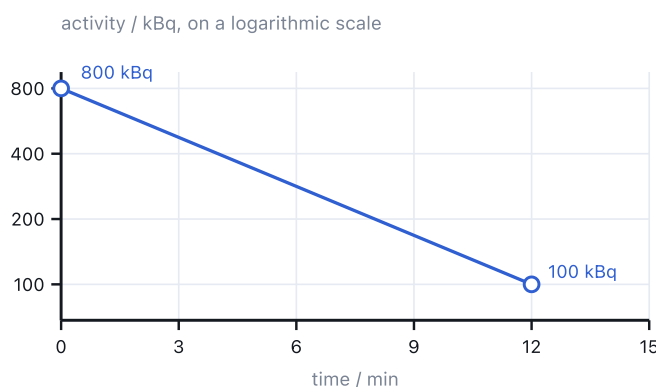
C $1/2$

D $4/9$

E $5/6$

Q11 · Half-life from a decay curve plotted with a logarithmic activity axis K

The activity of a pure radioactive source is measured over twelve minutes. The results are plotted with the activity on a **logarithmic** scale and the time on a linear scale.



The plotted points lie on a straight line. What is the half-life of the source?

A 1.5 min

B 3 min

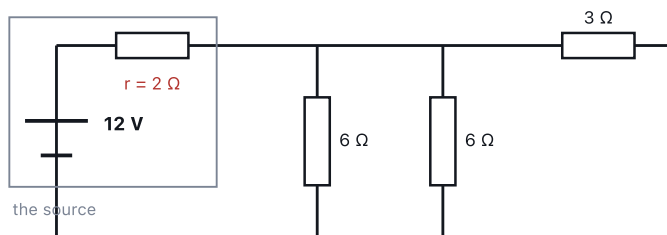
C 4 min

D 6 min

E 12 min

Q12 · Power in one resistor of a network driven by a cell with internal resistance H

A cell of electromotive force 12 V and internal resistance $2\ \Omega$ is connected to two $6\ \Omega$ resistors in parallel with each other, and that combination in series with a $3\ \Omega$ resistor.



The source has an internal resistance of $2\ \Omega$.

The two $6\ \Omega$ resistors are in parallel with each other.

What is the power dissipated in the $3\ \Omega$ resistor?

A 4.5 W

B 6.75 W

C 12 W

D 13.5 W

E 18 W

Q13 · The common speed after a perfectly inelastic collision between equal masses C

A trolley of mass 2.0 kg moving on a smooth horizontal track at 3.0 m s^{-1} collides with a second trolley of mass 2.0 kg that is at rest. The two trolleys stick together.

What is their common speed immediately after the collision?

A 0.75 m s^{-1}

B 1.5 m s^{-1}

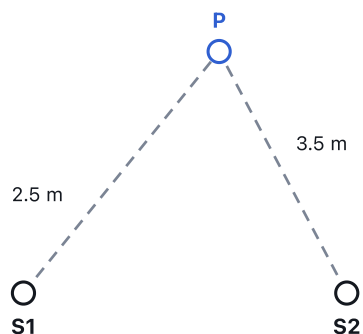
C 3.0 m s^{-1}

D 4.5 m s^{-1}

E 6.0 m s^{-1}

Q16 · Two-source interference: the lowest frequency for which a given point is a minimum **F**

Two loudspeakers **S1** and **S2** are driven by the same oscillator and therefore emit sound in phase. The speed of sound in air is 340 m s^{-1} . A microphone at the point **P** is 2.5 m from **S1** and 3.5 m from **S2**.



S1 and S2 are driven by the same oscillator, so they emit in phase.

What is the **lowest** frequency at which the sound heard at **P** is a minimum?

A 85 Hz

B 170 Hz

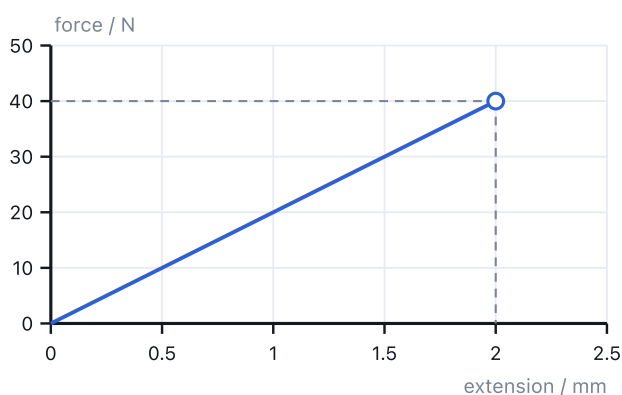
C 340 Hz

D 510 Hz

E 680 Hz

Q17 · The Young modulus from a force-extension graph and the dimensions of the wire **E**

The graph shows how the extension of a wire varies with the applied force, up to the limit of proportionality. The wire has an unstretched length of 2.0 m and a cross-sectional area of $1.0 \times 10^{-7} \text{ m}^2$.



What is the Young modulus of the material of the wire?

A $1.0 \times 10^{11} \text{ Pa}$

B $2.0 \times 10^{11} \text{ Pa}$

C $4.0 \times 10^{11} \text{ Pa}$

D $8.0 \times 10^{11} \text{ Pa}$

E $1.6 \times 10^{12} \text{ Pa}$

Q18 · How the number of photons emitted per second depends on power and wavelength **L**

A laser emits monochromatic light of wavelength λ at a power P . The wavelength is then **halved** and the power is **doubled**.

By what factor does the number of photons emitted per second change?

A $1/4$

B $1/2$

C 1

D 2

E 4

Q19 · The first-order approximation to a series combination when one term dominates **A**

Two springs of stiffness k and K are joined end to end, so that the pair is stretched by pulling on the two free ends. The combined stiffness is

$$k_{\text{combined}} = kK/(k + K)$$

In the case where K is very much larger than k , which expression is correct **to first order** in the small ratio k/K ?

A $k(1 - k/K)$

B $k(1 + k/K)$

C $k(1 - K/k)$

D $K(1 - k/K)$

E k

Q20 · The number of images of a point object between two plane mirrors at an angle **G**

Two plane mirrors are arranged with their reflecting surfaces facing each other, meeting along a common vertical edge at an angle of 60° . A small illuminated bead is placed between them, well away from the edge.

How many images of the bead are formed?

A 3

B 4

C 5

D 6

E 11

Q24 · An order-of-magnitude estimate of the mass of air in a room **A**

A classroom measures about **10 m** by **8 m** by **3 m**. The density of air is about **1.2 kg m^{-3}** .

Which value is the best estimate of the mass of the air in the room?

A **3 kg**

B **30 kg**

C **300 kg**

D **3000 kg**

E **30000 kg**

Q25 · The maximum kinetic energy of photoelectrons when the wavelength is halved **L**

The work function of a metal surface is **2.0 eV**. Light of wavelength **λ** incident on the surface produces photoelectrons whose maximum kinetic energy is **1.0 eV**.

The wavelength of the incident light is then halved. What is the new maximum kinetic energy of the photoelectrons?

A **1.0 eV**

B **2.0 eV**

C **3.0 eV**

D **4.0 eV**

E **6.0 eV**